SIH 2026 PROBLEM STATEMENT SIH26083

THERMOS

Thermal Health Risk & Early-Warning Management System

Prototype Version: 2.4.0 (Production)
Live App: thermo-safety.netlify.app
GitHub Repo: nagarjun012/THERMOS
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Executive Summary: THERMOS is a real-time biometeorological heat safety and disaster early-warning platform designed for citizens, outdoor laborers, and emergency authorities across India. By integrating multi-index environmental thermodynamics (UTCI, WBGT, Heat Index) into a single non-compensatory **Human Thermal Stress Score (HTSS)**, THERMOS eliminates false safety assurances caused by relying on air temperature alone.

1. PROBLEM STATEMENT & OPERATIONAL NEED

Standard weather forecasts report dry-bulb air temperature (T_a), which fails to communicate actual physiological heat strain on the human body. A dry air temperature of 40°C at 25% relative humidity is manageable through sweating, whereas 40°C at 70% relative humidity is lethal because atmospheric vapor pressure prevents sweat evaporation, driving core body temperature rapidly above 40.5°C (Heat Stroke).

Current System Limitations

- Reliance on single-variable temperature thresholds.
- Lack of occupational safeguards for outdoor workers.
- No localized physiological risk adjustments for elderly/sick.
- Stale daily advisories instead of hourly dynamic alerts.

THERMOS Breakthrough Solution

- Multi-Index Fused Human Thermal Stress Score (HTSS).
- Personal Safeguard Calculator with work-rest interval engine.
- City-wise GIS Heat Map & Disaster Command Center.
- Client-side zero-latency architecture using Open-Meteo API.

2. SIMPLIFIED & EASY-TO-UNDERSTAND FORMULA GUIDE

To make complex biometeorological thermodynamics intuitive for judges, officials, and citizens, THERMOS breaks calculations down into 4 clear steps:

1. Heat Index (HI) — What the air "Feels Like" with Humidity

Intuition: High humidity stops sweat from evaporating. When sweat cannot evaporate, your body traps heat internally.

```
Step 1: Calculate basic heat = 0.5 × (Temp + 61.0 + (Temp - 68)×1.2 + Humidity×0.094)
Step 2: If Temp ≥ 27°C, adjust for high moisture trapping using temperature-humidity multiplier.
```

2. Wet Bulb Globe Temperature (WBGT) — Outdoor Work Safety Index

Intuition: Used by military & industries to prevent heat stroke during physical labor outdoors. It adds direct sunlight heating to air moisture.

```
Vapor Pressure = (Humidity / 100) × 6.105 × e^((17.27 × Temp) / (237.7 + Temp)) Base
WBGT = 0.567 × Temp + 0.393 × Vapor Pressure + 3.94 Sunlight Bonus = If in direct
sun, add (Solar Radiation × 0.01) [Adds 3°C to 8°C under hot sun]
```

3. Universal Thermal Climate Index (UTCI) — Body Response Index

Intuition: Simulates how blood circulation, sweating, and skin temperature react to wind and sun.

```
Sun & Wind Adjustment = Air Temp + (0.08 × Solar Radiation) - (1.2 × √Wind Speed)
Final UTCI = Air Temp + 0.2×(Radiant Temp - Air Temp) - 0.1×Wind Speed +
0.05×Humidity
```

4. Human Thermal Stress Score (HTSS) — Fused Master Score (0 to 100)

Intuition: Combines all 3 indices with a **Safety Guardrail** so severe heat on any single index cannot be hidden.

```
Weighted Score = (45% × UTCI Score) + (35% × WBGT Score) + (20% × Heat Index Score)
Safety Guardrail = HTSS = Maximum of [Weighted Score, 0.85 × Highest Individual Sub-
Score]
```

THERMOS Technical Specification

HTSS Fusion Engine & System Architecture

3. HTSS RISK SPECTRUM & ACTION TABLE

HTSS RANGE	RISK LEVEL	PHYSIOLOGICAL IMPACT	REQUIRED ACTION & SAFEGUARD
0 – 30	SAFE	Normal thermoregulation balance.	Normal outdoor activities; maintain standard hydration.
31 – 50	LOW	Mild fatigue with prolonged exposure.	Hydrate 250ml/hour; monitor vulnerable individuals.
51 – 65	MODERATE	Heat cramps & exhaustion possible.	15-min rest per hour; mandatory water/ORS points.
66 – 80	HIGH	Heat exhaustion likely; stroke risk.	30-min shade rest per hour; halt heavy manual labor 12-3 PM.
81 – 100	EXTREME	Lethal heat stroke imminent.	Total outdoor labor shutdown; active emergency response.

4. SYSTEM ARCHITECTURE & TECHNICAL STACK

THERMOS is built as a zero-latency, client-side progressive application operating completely in-browser without server dependency, guaranteeing high availability during disaster situations.

Technology Stack

- **Frontend Logic:** Vanilla JavaScript (ES6+ Class Architecture).
- **Styling System:** CSS3 Glassmorphism with HSL ambient glowing mesh tokens.
- **GIS Mapping:** Leaflet.js 1.9.4 with Esri Dark Canvas basemap.
- **Analytics Charts:** Chart.js 4.4.0 (Line, Bar, Radar visualizations).
- **Telemetry API:** Open-Meteo REST API (Live 72-hour forecast).

Data Flow Architecture

1. User selects city or triggers GPS Geolocation API.
2. App executes asynchronous fetch to Open-Meteo REST endpoints.
3. Raw hourly arrays are piped into `ThermalStressEngine`.
4. Calculates HI, WBGT, UTCI, HTSS, and heatwave probability.
5. Re-renders Bento UI widgets, 3D icons, Leaflet map, and tables.

5. 3D DARK GLASSMORPHISM WEATHER DASHBOARD DESIGN

The dashboard implements a modern **Dark Glassmorphism Bento UI** design with vibrant ambient background glows (Indigo, Violet, Cyan) and 3D weather icons that dynamically switch based on live telemetry:

3D Sun Icon	3D Rain Cloud	3D Storm Cloud

Triggered when $T \ge 38^\circ\text{C}$. Represents sunny heatwave conditions.	Triggered when Rain Codes or $RH > 85\%$. Translucent water drops animation.	Triggered for Extreme Risk or $T \ge 44^\circ\text{C}$. Dark lightning elements.

THERMOS Interactive Modules

Personal Safeguards, Government Command & Verification

6. PERSONAL HEAT RISK SAFEGUARD CALCULATOR

The Personal Risk Calculator provides individualized heat strain evaluation based on physiological and workload factors. It modifies baseline HTSS using validated medical vulnerability multipliers:

VULNERABILITY FACTOR	INPUT OPTIONS	HTSS RISK OFFSET ADJUSTMENT
Age Group	Under 18 / 18–50 / 51–65 / Over 65	+5 to +12 points for elderly / children.
Work Intensity	Sedentary / Moderate / Heavy Manual Labor	+0 to +18 points for heavy labor outdoors.
Hydration Status	Optimal / Moderate / Dehydrated	+10 points for poor fluid intake.
Acclimatization	Acclimatized (>14 days) / Unacclimatized	+8 points for unacclimatized outdoor exposure.
Health Conditions	Cardiovascular, Diabetes, Respiratory, Kidney	+6 points per active medical condition.

Calculated Action Outputs: The calculator outputs exact personalized guidelines:

- **Max Safe Outdoor Duration:** Continuous limit (e.g. 45 minutes before forced break).
- **Hydration Rate:** Recommended fluid consumption (e.g. 0.85 Liters/hour ORS/water).
- **Mandatory Rest Interval:** Shadow cooling break requirement (e.g. 15 mins rest per 45 mins work).

7. GOVERNMENT COMMAND CENTER & DISASTER MANAGEMENT MATRIX

Designed for District Magistrates, State Disaster Management Authorities (SDMAs), and NDMA emergency response units:

- **City-wise Risk Ranking Table:** Real-time table sorting 35 major Indian cities across plains, coastal, and hilly regions by HTSS, population exposure, and alert level.
- **State Vulnerability Assessment:** Multi-factor radar metrics evaluating Elderly %, Outdoor Worker %, Poverty Density, and Healthcare Capacity Deficit.
- **Hospital Load Projections:** Predictive 3-5 day hospitalization rate index (per 100k population) enabling proactive medical supply allocation.

8. PRODUCTION READINESS & VERIFICATION SUMMARY

Live Website URL	https://thermo-safety.netlify.app/
GitHub Repository	https://github.com/nagarjun012/THERMOS.git
Codebase Quality	100% Client-Side JS, zero syntax errors, verified via Node & Playwright.

Performance	First Contentful Paint < 0.8s, Lightweight CDN distribution.